

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the particle by forming a second order differential equation. (must have correct terms but allow sign errors)	AO3.3	M1	$M \frac{d^2 x}{dt^2} = -4M \frac{dx}{dt} - 8Mx$ $\frac{d^2 x}{dt^2} + 4 \frac{dx}{dt} + 8x = 0$
	Obtains correct differential equation	AO1.1b	A1	$\lambda^2 + 4\lambda + 8 = 0$ $\lambda = -2 \pm 2i$
	Forms and solves auxiliary equation for 'their' D.E.	AO1.1a	M1	Complex roots $\Rightarrow$ General solution is of the form:
	States a correct form of the general solution for 'their' auxiliary solution. (ft only if both M1 marks have been awarded)	AO1.1b	A1F	$x = Ae^{-2t} \cos(2t + B)$ $\dot{x} = -2Ae^{-2t} \cos(2t + B)$ $- 2Ae^{-2t} \sin(2t + B)$
	Uses initial conditions to find arbitrary constants for 'their' solution	AO1.1a	M1	$\dot{x}(0) = 0 \text{ so,}$ $-2A \cos(B) - 2A \sin(B) = 0$
	Obtains correct value for one of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$\Rightarrow \tan B = -1$ $\Rightarrow B = -\frac{\pi}{4}$
	Obtains correct value for both of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$x(0) = 1$ $\Rightarrow \frac{A}{\sqrt{2}} = 1$
	Uses 'their' model to describe the motion of the particle either as a written description or shown on a clearly labelled graph.	AO3.4	A1F	$\Rightarrow A = \sqrt{2}$ $x = \sqrt{2}e^{-2t} \cos\left(2t - \frac{\pi}{4}\right)$ $\therefore$ the particle oscillates about O, with period π seconds and decreasing amplitude.
(b)	Refines their DE model to account for altered resistive force by introducing a new	AO3.5c	B1	$\frac{d^2 x}{dt^2} + \alpha \frac{dx}{dt} + 8x = 0$

coefficient for $\frac{dx}{dt}$			Critical damping $\Rightarrow$
Uses or states condition for critical damping	AO1.2	B1	$\lambda^2 + \alpha\lambda + 8 = 0$ must have equal roots
Deduces value for $\frac{dx}{dt}$ coefficient of	AO2.2a	R1	$\alpha^2 = 32$ $\alpha = 4\sqrt{2}$
States resistive force	AO3.4	B1	Resistive force should have magnitude $4\sqrt{2}Mv$
Total 12 marks			

Q2.

(a) (i) $m\ddot{x} = -\frac{\lambda x}{a}$ M1

$= -amn^2 \frac{x}{a}$

$\ddot{x} = -n^2 x$ SHM A1

(ii) $T = \frac{2\pi}{n}$ 2

B1

1

(b) (i) $m\ddot{x} = -amn^2 \frac{x}{a} - mkv$ M1

A1

$\ddot{x} + k\dot{x} + n^2x = 0$

AG A1

3

(ii) $p^2 + \frac{5n}{2}p + n^2 = 0$

$\left(p + \frac{5n}{4}\right)^2 - \frac{9n^2}{16} = 0$ M1

$p + \frac{5n}{4} = \pm \frac{3n}{4}, \quad p = -2n, \quad p = -\frac{n}{2}$



$$2p^2 + 5np + 2n^2 = 0$$

$$(2p + n)(p + 2n) = 0$$

$$p = -\frac{n}{2}, p = -2n$$

A1

$$x = Ae^{-2nt} + Be^{-\frac{n}{2}t}$$

M1

$$t = 0, x = 0: A + B = 0$$

$$t = 0, \dot{x} = U$$

$$\dot{x} = -2nAe^{-2nt} - \frac{n}{2}Be^{-\frac{n}{2}t}$$

m1

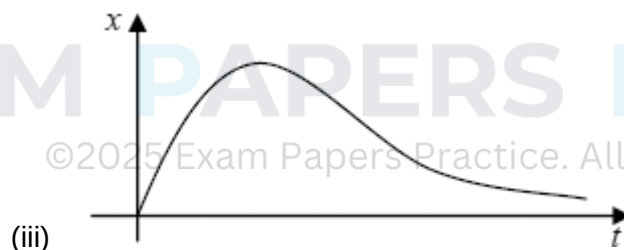
$$U = -2nA - \frac{n}{2}B$$

$$A = -\frac{2U}{3n} \quad B = \frac{2U}{3n}$$

$$x = \frac{2U}{3n} \left(e^{-\frac{nt}{2}} - e^{-2nt} \right)$$

A1,A1

6



Accept sketch with correct shape not reaching origin
but not crossing x -axis elsewhere

Accept reference to real distinct roots of auxiliary
equation

B1

Heavy damping

Independent of previous mark

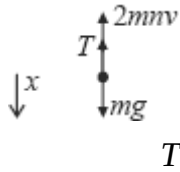
B1

2

[14]

Q3.

(a)



B1

$$m\ddot{x} = mg - \frac{5mn^2a}{a} \left(x + \frac{g}{5n^2} \right) - 2mn\dot{x}$$

all terms

M1A1A1

$$\ddot{x} + 2n\dot{x} + 5n^2x = 0$$

CAO; AG

A1

5

(b) AE: $p^2 + 2np + 5n^2 = 0$

M1

$$(p + n)^2 + 4n^2 = 0$$

$$p = -n \pm 2ni$$

A1

$$x = e^{-nt} (A \cos 2nt + B \sin 2nt)$$

A1

$$t = 0, x = 0 \Rightarrow A = 0$$

A1

$$\dot{x} = -e^{-nt} (B \sin 2nt) + e^{-nt} (2nB \cos 2nt)$$

M1

$$t = 0, x = U \Rightarrow$$

$$U = 2nB \quad B = \frac{U}{2n}$$

A1

$$x = \frac{U}{2n} e^{-nt} \sin 2nt$$

A1

7

(c) light

B1

1

(d) $x = 0 \therefore \sin 2nt = 0$

M1

$$2nt = (0, \dots) \pi, \dots$$

$$t = \frac{\pi}{2n}$$

A1

$$\dot{x} = -e^{-nt} \left(\frac{U}{2n} \sin 2nt \right) + e^{-nt} (U \cos 2nt)$$

$$= -e^{-\frac{\pi}{2}} (0) + e^{-\frac{\pi}{2}} (U)(-1)$$

M1

$$\text{speed} = Ue^{-\frac{\pi}{2}}$$

A1

4

[17]

Q4.

(a) Natural length of AP is $4a$ and natural length of BP is $2a$
When particle is x from equilibrium position:

$$\frac{4mn^2 a(2a+x)}{4a}$$

Tension in AP is

M1A1

$$\frac{4mn^2 a(a-x)}{2a}$$

Tension in BP is

M1A1

In general position, using $F = ma$:

$$m \frac{d^2 x}{dt^2} = \frac{4mn^2(a-x)}{2a} - \frac{4mn^2 a(2a+x)}{4a} - 2mn \frac{dx}{dt}$$

M1A1

$$m\ddot{x} = 2mn^2 a - 2mn^2 x - 2mn^2 a - mn^2 x - 2mn\dot{x}$$

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 3n^2 x = 0$$

AG

A1

7

- (b) [Substituting $x = Ae^{pt}$]
 $p^2 + 2p + 3 = 0$

M1A1

$$p = -1 \pm \sqrt{2}i$$

A1

General solution is:

$$x = e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t)$$

A1

When $t = 0$, $x = \frac{1}{2}a \Rightarrow \frac{1}{2}a = A$

B1

Differentiating:

$$\frac{dx}{dt} = -e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t) + e^{-t}(-A\sqrt{2} \sin \sqrt{2}t + B\sqrt{2} \cos \sqrt{2}t)$$

M1A1

When $t = 0$, $\frac{dx}{dt} = 0$
 $\Rightarrow 0 = -A + \sqrt{2}B$

$$A = \frac{1}{2}a, B = \frac{1}{2\sqrt{2}}a$$

A1

$$x = ae^{-t}\left(\frac{1}{2} \cos \sqrt{2}t + \frac{1}{2\sqrt{2}} \sin \sqrt{2}t\right)$$

8

[15]

Q5.

- (a) Using $F = ma$,

$$2m\ddot{x} = 2mg - T - 4mnv$$

3 terms at least correct

M1

$$= 2mg - 8mn^2a \cdot \frac{x}{a} - 4mnv$$

A1

$$m\ddot{x} + 4mn^2x + 2mn\dot{x} = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 4n^2x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

$$p^2 + 2np + 4n^2 = 0$$

M1

$$p = \frac{-2n \pm \sqrt{4n^2 - 16n^2}}{2}$$

$$= -n \pm \sqrt{3}in$$

A1

C.F. is

$$x = e^{-nt}(A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$$

Dep M1

B1ft

$$\text{P.I. is } x = \frac{g}{4n^2}$$

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B1

General solution is $x =$

$$e^{-nt}(A \cos \sqrt{3}nt + B \sin \sqrt{3}nt) + \frac{g}{4n^2}$$

ft dep M1 first line (b)

M1

At least one of CF, PI correct

A1ft

$$\text{When } t = 0, x = 0, \Rightarrow 0 = A + \frac{g}{4n^2}$$

M1

$$\therefore A = -\frac{g}{4n^2}$$

A1

When $t = 0$, $\frac{dx}{dt} = 0$;

$$\frac{dx}{dt} = -ne^{-nt}(A\cos\sqrt{3}t + B\sin\sqrt{3}t) +$$

M1

$$e^{-nt}(-\sqrt{3}nA\sin\sqrt{3}nt + \sqrt{3}nB\cos\sqrt{3}nt)$$

A1

$$\Rightarrow 0 = -nA + \sqrt{3}nB$$

$$\therefore B = \frac{1}{\sqrt{3}}A = -\frac{g}{4\sqrt{3}n^2}$$

A1

$$\therefore x = \frac{g}{4n^2} - \left(\frac{g}{4n^2} \cos\sqrt{3}nt + \frac{g}{4\sqrt{3}n^2} \sin\sqrt{3}nt \right) e^{-nt}$$

A1

(c) $\frac{dx}{dt} = ne^{-nt} \left(\frac{g}{4n^2} \cos\sqrt{3}nt + \frac{g}{4\sqrt{3}n^2} \sin\sqrt{3}nt \right)$

M1

$$-e^{-nt} \left(-\frac{\sqrt{3}g}{4n^2} \sin\sqrt{3}nt + \frac{g}{4n} \cos\sqrt{3}nt \right)$$

A1

$$\frac{dx}{dt} = 0 \text{ when}$$

$$\cos\sqrt{3}nt + \frac{1}{\sqrt{3}} \sin\sqrt{3}nt + \sqrt{3} \sin\sqrt{3}nt$$

$$- \cos\sqrt{3}nt = 0$$

M1

$$\sin \sqrt{3} \pi t = 0$$

$$\text{First time at rest is when } t = \frac{\pi}{\sqrt{3}\pi}$$

A1

4

[19]

Q6.

(a) $T = 10(x - 0.5)$ or $-10(x - 0.5)$

M1A1

2

(b) $10(x - 0.5) - 2v = 1 \frac{dv}{dt}$

Newton's 2nd law

M1A1F

$$\frac{dv}{dt} + 2v - 10x = -5$$

AG

A1

3

- (c) (i) The rate of change of the length of the spring = speed of A relative to B

OE

E1

1

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$$\therefore \dot{x} = 1 - v$$

(ii) $-\ddot{x} + 2(1 - \dot{x}) - 10x = -5$

M1

$$\ddot{x} + 2\dot{x} + 10x = 7$$

AG

A1

2

(d) Auxiliary equation: $m^2 + 2m + 10 = 0$

M1

$$m = -1 \pm 3i$$

A1

CF $x = e^{-t} (A \sin 3t + B \cos 3t)$

M1

PI $x = \alpha$

$\dot{x} = \ddot{x} = 0$

M1

A1F

$\alpha = \frac{7}{10}$

GS $x = e^{-t} (A \sin 3t + B \cos 3t) + \frac{7}{10}$

A1F

6

(e) $t \rightarrow \infty$

$e^{-t} \rightarrow 0$

$x \rightarrow \frac{7}{10}$

M1A1F

2

EXAM PAPERS PRACTICE [16]

Q7. ©2025 Exam Papers Practice. All rights reserved.

(a) When displacement is x ,

Tension $= \frac{\lambda x}{l} = \frac{4mn^2 a x}{2a}$

$= 2mn^2 x$

B1

Using $F = ma$,

$m = \frac{d^2 x}{dt^2} = mg - 2mn^2 x - 2mn \frac{dx}{dt}$

M1 A1

$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + 2n^2 x = g$

A1

4

(b) CF $x = Ae^{pt}$
 $p^2 + 2np + 2n^2 = 0$

M1

$$p = \frac{-2n \pm \sqrt{4n^2 - 8n^2}}{2}$$

$$= -n \pm ni$$

A1

$$\therefore x = e^{-nt}(A \cos nt + B \sin nt)$$

A1

$$\text{PI } x = \frac{g}{2n^2}$$

B1

$$x = e^{-nt}(A \cos nt + B \sin nt) + \frac{g}{2n^2}$$

M1

When $t = 0$, $x = 0$,

$$A = -\frac{g}{2n^2}$$

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B1

$$\frac{dx}{dt} = -ne^{-nt}(A \cos nt + B \sin nt) +$$

M1

$$e^{-nt}(-nA \sin nt + nB \cos nt)$$

When $t = 0$, $\frac{dx}{dt} = 0$

$$\Rightarrow -nA + nB = 0$$

$$B = \frac{g}{2n^2}$$

A1

$$x = \frac{g}{2n^2} \{1 - e^{-nt} (\cos nt + \sin nt)\}$$

A1

9

[13]

Q8.

(a) $T_{AP} = \frac{\lambda \cdot 3a}{4a} = \frac{3}{4} \lambda$

$$T_{PB} = 4mg \cdot \frac{a}{2a} = 2mg$$

Either

B1

Using $F = ma$ vertically

M1

$$mg + T_{PB} = T_{AP}$$

$$\therefore mg + 2mg = \frac{3}{4} \lambda$$

A1

$$\lambda = 4mg$$

A1

4

- (b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

$$mg, T_{AP} = \frac{\lambda(3a+x)}{4a} = \frac{mg(3a+x)}{a},$$

M1

$$T_{PB} = 4mg \cdot \frac{(a-x)}{2a} = \frac{2mg}{a}(a-x)$$

and resistance $\frac{1}{3}mk\dot{x}$
All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{3}mk\dot{x}$$

Dependent on both M1 above

m1

$$m\ddot{x} = mg + \frac{2mg}{a}(a-x) - \frac{mg(3a+x)}{a} - \frac{1}{3}mk\dot{x}$$

A1

$$\ddot{x} - g - \frac{2g}{a}(a-x) + \frac{g(3a+x)}{a} + \frac{1}{3}k\dot{x} = 0$$

$$\ddot{x} + \frac{1}{3}k\dot{x} + \frac{3gx}{a} = 0$$

A1

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

A1

6

(ii) Substituting $x = Ae^{nt}$,

$$10n^2 + 2kn + 5k^2 = 0$$

$$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$$

M1

$$= \frac{1}{10}(-k \pm 7ki)$$

A1

$$x = e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt)$$

M1 A1ft

$$\text{When } t = 0, x = \frac{a}{2}, A = \frac{a}{2}$$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{k}{10}e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt) +$$

$$e^{-\frac{k}{10}t} \left(-\frac{7}{10}kA \sin \frac{7}{10}kt + \frac{7}{10}kB \cos \frac{7}{10}kt \right)$$

M1 A1ft

$$\text{When } t = 0, \frac{dx}{dt} = 0, 0 = \frac{k}{10}A + \frac{7}{10}kB$$

M1

$$B = \frac{\alpha}{14}$$

$$x = \frac{\alpha}{14} e^{-\frac{k}{10}t} (7 \cos \frac{7}{10}kt + \sin \frac{7}{10}kt)$$

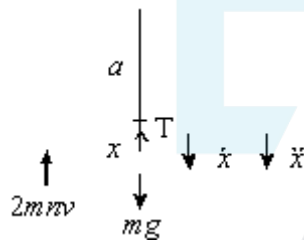
A1

9

[19]

Q9.

(a)



Using $F = ma$,

$$m\ddot{x} = mg - T - 2mnv$$

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M1

$$= mg - 6mn^2 \frac{x}{\alpha} - 2mnv$$

A1

$$m\ddot{x} + 6mn^2 x + 2mnv = mg$$

$$\text{i.e. } \frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 6n^2 x = g$$

A1

3

(b) Substituting $x = Ae^{pt}$

M1

$$p^2 + 2np + 6n^2 = 0$$

$$p = \frac{(-2n \pm \sqrt{4n^2 - 24n^2})}{2}$$

$$= -n \pm \sqrt{5} i n$$

A1

$$\therefore \text{CF is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

A1

$$\text{PI is } x = \frac{g}{6n^2}$$

B1

$$\therefore \text{GS is } x = e^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

$$= \frac{g}{6n^2}$$

A1

$$\text{When } t = 0, x = 0 \Rightarrow 0 = A + \frac{g}{6n^2}$$

M1

$$\therefore A = -\frac{g}{6n^2}$$

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A1

$$\text{When } t = 0, \frac{dx}{dt} = 0;$$

$$\frac{dx}{dt} = -ne^{-nt} (A \cos \sqrt{5} nt + B \sin \sqrt{5} nt)$$

$$+ e^{-nt} (-\sqrt{5} n A \sin \sqrt{5} nt + \sqrt{5} n B \cos \sqrt{5} nt)$$

$$+ e^{-nt} (-\sqrt{5} n A \sin \sqrt{5} nt + \sqrt{5} n B \cos \sqrt{5} nt)$$

M1

$$\Rightarrow 0 = -n A \times \sqrt{5} n B$$

$$\therefore B = \frac{1}{\sqrt{5}} A = -\frac{g}{6\sqrt{5}n^2}$$

A1

$$\therefore x = \frac{g}{6n^2} -$$

$$\left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) e^{-nt}$$

A1

10

(c) $\frac{dx}{dt} =$

$$ne^{-nt} \left(\frac{g}{6n^2} \cos \sqrt{5}nt + \frac{g}{6\sqrt{5}n^2} \sin \sqrt{5}nt \right) -$$

$$e^{-nt} \left(-\frac{\sqrt{5}g}{6n} \sin \sqrt{5}nt + \frac{g}{6n} \cos \sqrt{5}nt \right)$$

M1

$$\frac{dx}{dt} = 0 \text{ when } \cos \sqrt{5}nt + \frac{1}{\sqrt{5}} \sin \sqrt{5}nt +$$

$$\sqrt{5} \sin \sqrt{5}nt \cos \sqrt{5}nt = 0$$

$$\sin \sqrt{5}nt = 0$$

M1A1

First time at rest is when $t = \frac{\pi}{\sqrt{5}n}$

$$\text{When } t = \frac{\pi}{\sqrt{5}n}, \quad \frac{dx}{dt} = 0 \quad \text{SC2}$$

A1

4

[17]

Examiner reports

Q2.

Candidates were able to attempt all parts of this question, but part (a) proved more challenging than expected, with some seeming unsure as to the requirements for simple harmonic motion. Part (b)(i) was done very well, as was part (b)(ii), although there were a number of errors in accuracy here. Part (b)(iii) proved very challenging; many successfully quoted the nature of the roots of the auxiliary equation as justification for heavy damping, but the hint of using a sketch was not appreciated.

Q3.

Part (a) was answered well, with occasional errors in finding the extension of the string. Part (b) was less successful, with some only being able to attempt the solving of the auxiliary equation before stopping. There were various algebraic errors, the most frequent being the omission of 'n' at varying stages, but still a number of good solutions. The answer to part (c) was well known. Part (d) proved quite testing in choosing correct methods for solution, and again algebraic errors marred solutions.

Q4.

Weaker candidates created the printed differential equation in part (a) by dubious means. It was necessary to find the equilibrium position and then use the extensions to be $2a + x$ in string AP and $a - x$ in string BP . Part (b) was answered very well by most candidates.

The only common error was in trying to insert the boundary conditions and finding $\frac{dx}{dt}$, when $\frac{d}{dx} \cos \sqrt{2}t$ sometimes became $-\sin \sqrt{2}t$ rather than $-\sqrt{2} \sin \sqrt{2}t$.

Q5.

Part (a) was answered well, although again some candidates were clearly working back from the printed result to obtain the initial equation of motion. Part (b) was attempted well while the common error was in using $x = a$ when $t = 0$, not noticing that x was the distance below A , and initially the particle was released at A ; thus $\alpha = 0$ when $t = 0$. Relatively few candidates were totally accurate in finding the velocity of the particle in part (c) and only these successful candidates could notice that the terms in $\cos \sqrt{3\pi t}$ cancelled out, leaving only the terms containing $\sin \sqrt{3\pi t}$.

Q6.

The great majority of candidates were able to use Hooke's law and Newton's second law to answer parts (a) and (b) of this question. Some candidates were unable to explain why $\dot{x} = 1 - v$ for part (c)(i) but were still able to gain full marks for part (c)(ii).

Some candidates were able to answer part (d) of the question without answering part (c). Even though only the general solution for the differential equation was requested, some candidates proceeded to find the particular solution also. Many candidates were able to state the limiting value of x requested in part (e).

Q7.

In part (a), a significant proportion of candidates worked back from the given result. Others used x as the position beyond the equilibrium position rather than as stated, and then had to 'lose' terms. In part (b), candidates made very good progress with many obtaining the correct result. The common error was forgetting an " n " in the

differentiation of one of the terms when finding $\frac{dx}{dt}$ to insert a boundary condition.

Other candidates could not simplify correctly the term $\frac{-2n \pm \sqrt{-4n^2}}{2}$.

Q8.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB , with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q9.

Many candidates attempted to find the extension of the string in the equilibrium position, although in this question it was not relevant. These candidates had no mg force present when they found their differential equation. As this equation was printed on the question paper, candidates who then 'created' the mg force rarely scored marks in part (a). In part (b), candidates usually obtained the general solution $x = e^{-nt} (A \cos \sqrt{5}nt + B \sin \sqrt{5}nt)$, and most attempted to find the particular solution. They then showed the correct method for obtaining the actual solution. Those who correctly found the answer to part (b) often

substituted $t = \frac{\pi}{\sqrt{5}n}$ to show that the particle was at rest.

Only the better candidates differentiated their answer to show that the particle was at rest.